

Hassan EL QADI

Finance and Decision-Making Engineer | Open to Opportunities

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in el-qadi

🌐 hassanelq

🌐 Portfolio

Education

National School of Applied Sciences (ENSA) – Agadir

2023 – 2026

Engineering Degree – Finance and Decision-making Engineering

- Stochastic calculus, Financial mathematics, Optimization, Stochastic processes and Monte Carlo simulation, Pricing of financial products, Derivatives and yield curves, Numerical methods in finance, Portfolio and risk management, Econometrics and time series, Advanced programming (C++, Python), Data Science and Big Data.

National School of Applied Sciences (ENSA) – Agadir

2021 – 2023

Preparatory Cycle

- Algebra, Analysis, Probability and Statistics, Ordinary Differential Equations.

Professional Experience

Finamaze

Feb – Jul 2026

Intern – Quantitative Developer (Final-Year Project)

- Designed and developed a multi-asset stress-testing framework using Entropy Pooling and copulas, integrating crisis scenarios and measuring their impact on portfolio risk metrics (VaR, CVaR, drawdowns).
- Contributed to backend development and monitoring for an investment application deployed for Saudi Arab Bank.
- Collected, validated, and maintained financial market data from multiple sources.

Oracle Capital

Feb – Jun 2025

Intern – Quantitative Finance / Data Science

- Calibrated the Heston stochastic volatility model for option pricing using implied volatility surfaces.
- Compared three calibration approaches: classical optimization, Fast Fourier Transform (FFT), and deep learning.
- Analyzed vanilla options on the SPX index to optimize model parameters and assess the quality of model fit.
- Evaluated accuracy and computation time for each method: 2,650s (closed-form), 45s (FFT), 0.12s (DL).

Academic Projects

European & American Option Pricing | *Python, FastAPI*



- Implemented Black-Scholes closed-form pricing and Monte Carlo simulation with a stochastic Euler scheme.
- Implemented binomial and trinomial trees, analyzing the early-exercise boundary and convergence of option values.

Interest Rate Swap Pricing | *Python, Flask*



- Priced fixed-for-floating interest rate swaps using market zero-coupon yield curves.
- Implemented and calibrated Vasicek and CIR short-rate models to model interest rates and reproduce the yield curve.

Portfolio Optimization | *Python, Streamlit*



- Implemented several asset allocation models: Mean-Variance, Global Minimum Variance, Risk Parity, and Black-Litterman.
- Ran out-of-sample backtests to evaluate portfolio performance, volatility, drawdown, and allocation stability.

Financial Market Sentiment Analysis | *Next.js, Python, FastAPI*



- Deployed FinBERT, a pre-trained NLP model, to classify sentiment in financial content from multiple sources (Twitter, Reddit, Finviz) and generated short-term market signals from aggregated sentiment scores.

Skills and Languages

Programming: Python, C++, C, Java, R, JavaScript, VBA.

Data Science & Machine Learning: Pandas, NumPy, Scikit-Learn, TensorFlow, ETL, web scraping, NLP.

Backend, APIs & Tools: FastAPI, Flask, Git, GitHub, Docker, CI/CD, SQL, PostgreSQL.

Languages: English (professional), French (professional), Arabic (native).

Certifications

Financial Markets (Yale University | Coursera)



Overview of capital markets, corporate finance, risk management, and behavioral finance.

Markets Quantitative Analysis (MQA) Job Simulation (Citi | Forage)

Quantitative Research Job Simulation (JP Morgan | Forage)

Volunteering

Financial day – ENSA Agadir (4th & 5th ed.): Digital strategy, media content, and event organization.

Association Des Étudiants – ENSA Agadir (2023–2024) : Student event organization, communication, and media.